



POLITECNICO
MILANO 1863



Aerospace technologies and materials

Additive Manufacturing project

Academic year: 2023/2024

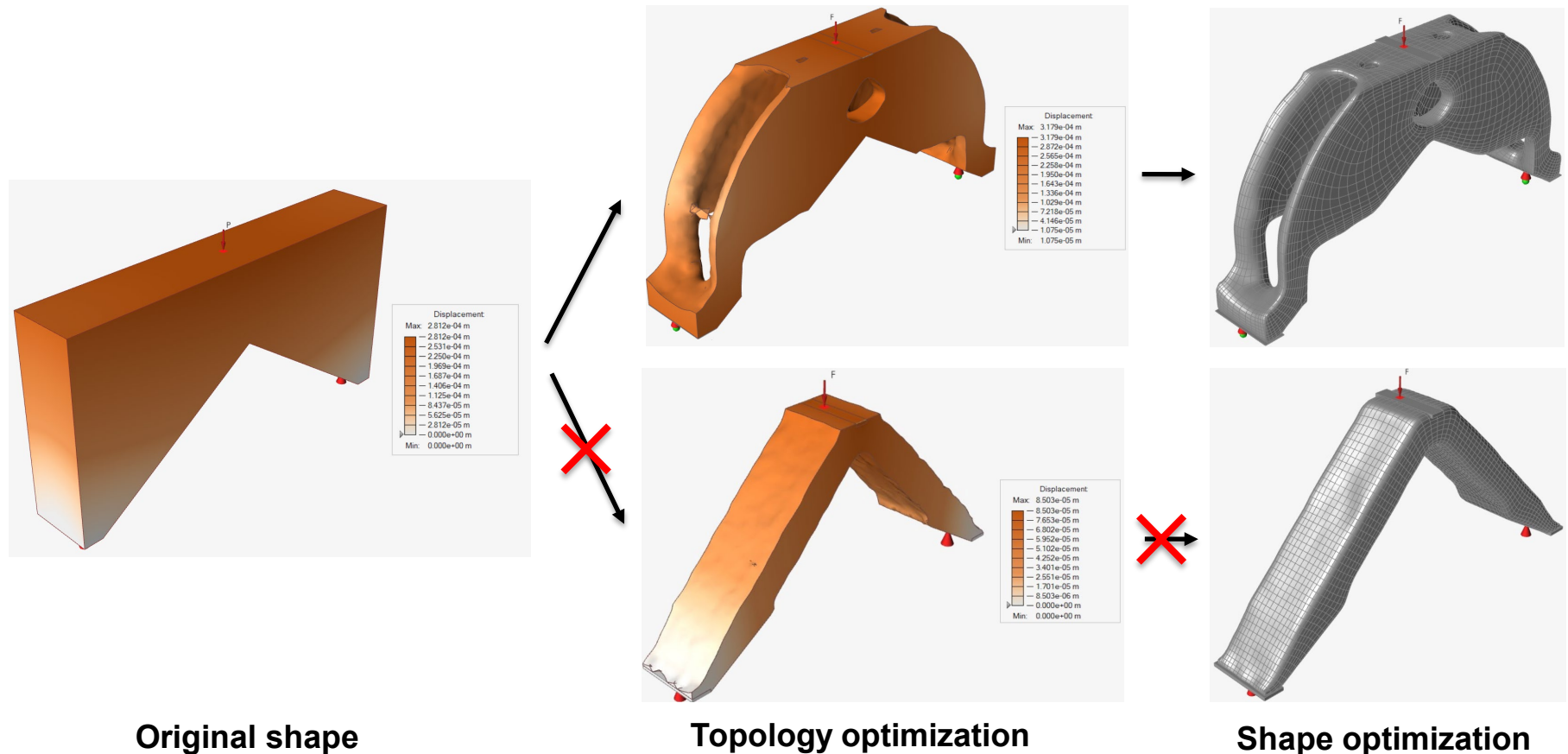
Group 8

Students: Diego Loras, Daniel Cano, David Gonzalez, Edoardo Paolini, Oscar Pratola, Giulia Leda Spinelli, Matteo Verri

Optimization

- **Goal:** maximizing $\frac{1}{m \cdot \delta z} [g^{-1} \cdot mm^{-1}]$
- **Softwares:** SolidWorks and Altair Inspire

The following configurations have been considered:



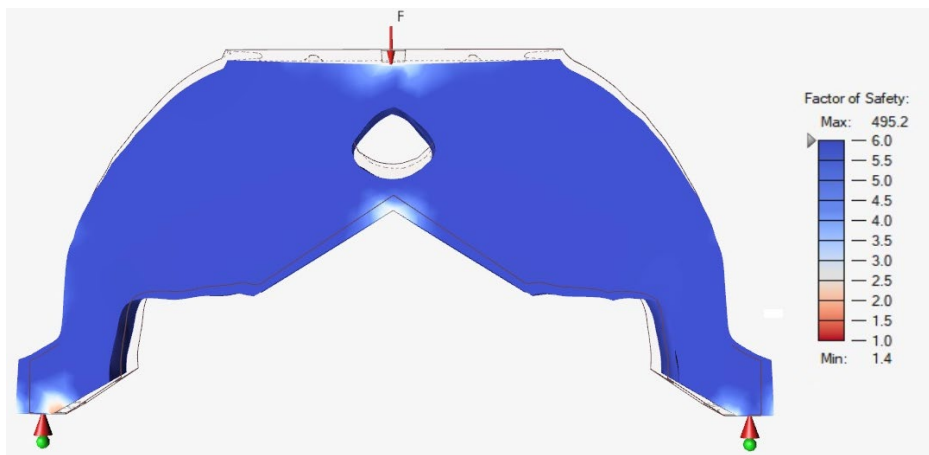
Process

- **Test configuration:** uniform distributed load (2 mm·10 mm·150 N)
- **Constraints:** fixed support (V-shape), simply support (Ω-shape)
- **Simulations:** target mass 36% of the total one. Goal: maximizing $\frac{1}{m \cdot \delta z}$
- **Printing orientation choice:** horizontal to reduce supports and printing time.
Furthermore, this allows to have the fibers aligned to the load applied
- **Discarded choices:** V-shape (prone to bend with excessive vertical displacement) and vertical printing: unnecessary waste of material for supports
- **Chosen configuration:** Ω-shape, as it shows a better mechanical behaviour with a limited displacement

Results

Model	Weight [%]	Vertical displ. [mm]	Printing time [min]	$\frac{1}{m \cdot \delta z} [g^{-1} \cdot m^{-1}]$
Original	100 (i.e. 14.18 g)	0.173	17	406.7
Ω -shape I	36 (i.e. 5.17 g)	0.291	55	666.2
Ω -shape II	25	0.409	/	684.4

➤ Ω -shape II surpasses yielding point, so **Ω -shape I** is the chosen design

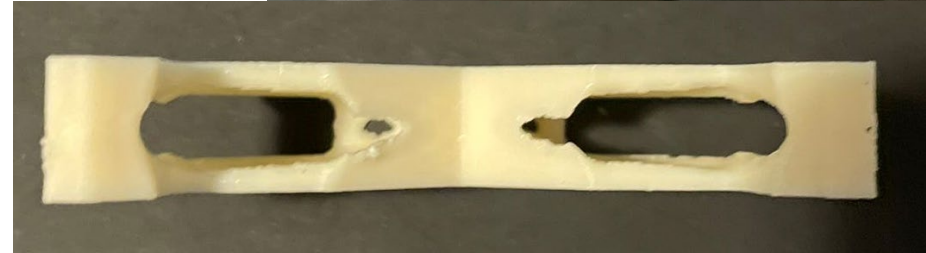
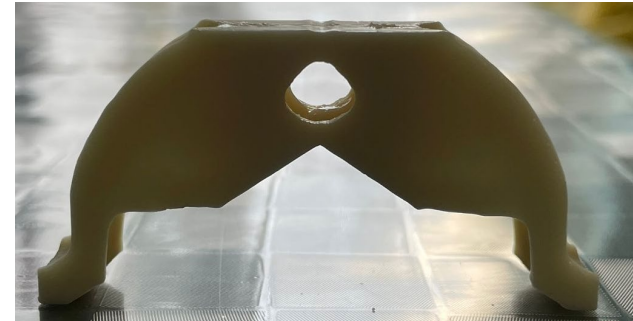
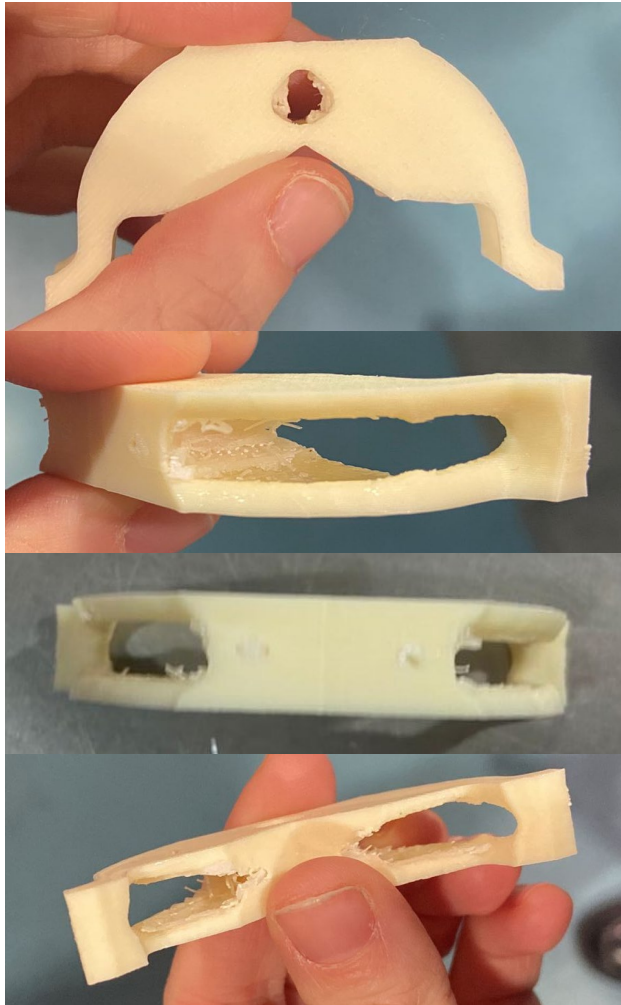


a) Ω -shape I front view factor of safety



b) Ω -shape I bottom view factor of safety

Post Manufacturing: *Support removal with NaOH 22g/L solution*



Test Results



	Mass [g]	Displacement [mm]	$\frac{1}{m \cdot \delta z} [g \cdot m]^{-1}$
Estimated	5,17	0,2910	666,2
Printed	4,76	0,5594	375,6
Support Removal	<4,76		

Testing data

